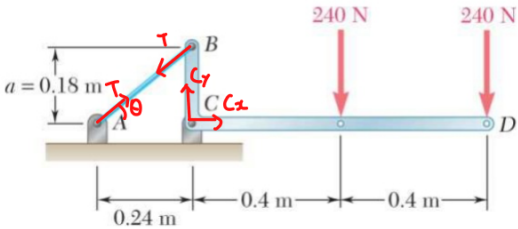


Problem 4.19: The bracket BCD is hinged at C and attached to a control cable at B . For the loading shown, determine (a) the tension in the cable, (b) the reaction at C .



$$\theta = \tan^{-1}(0.18/0.24)$$

$$= 36.869^\circ$$

system BCD on cable

$$\sum F_x = -T \cos \theta + C_x = 0$$

$$\sum F_y = -T \sin \theta + C_y - 240 - 240 = 0$$

$$\sum M_C = 0.18 \times T \cos \theta - 0.4 \times 240 - 0.8 \times 240 = 0$$

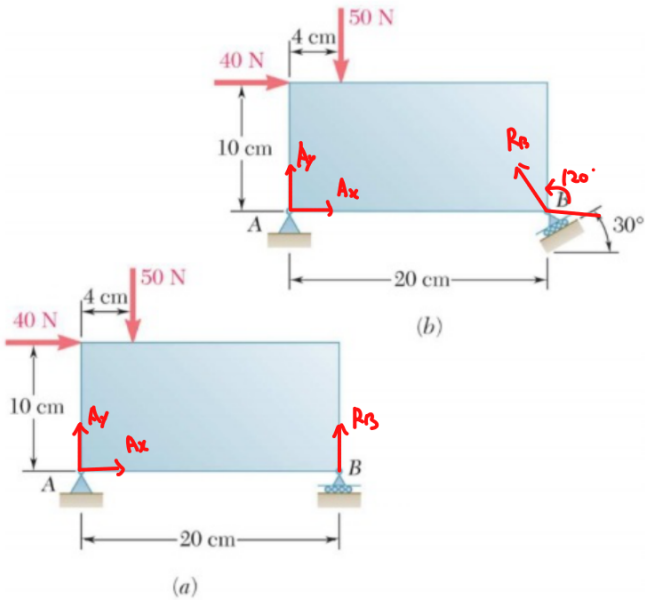
$$a) \quad T = 2000 \text{ N}$$

$$\therefore C_x = 1600 \text{ N}$$

$$C_y = 1680 \text{ N}$$

$$b) \quad C = \sqrt{C_x^2 + C_y^2} = 2320 \text{ N} \quad \angle 46.4^\circ$$

Problem 4.23: For each of the plates and loadings shown, determine the reactions at A and B .



a)

$$\sum F_x = 0$$

$$A_x + 40 = 0$$

$$\therefore A_x = -40$$

$$\sum F_y = 0$$

$$A_y + R_B - 50 = 0$$

$$\sum M_A = 0$$

$$-0.1 \times 40 - 0.04 \times 50 + 0.2 \times R_B = 0$$

$$R_B = 30 \text{ N}$$

$$\therefore A_y = 20 \text{ N}$$

$$A = \sqrt{A_x^2 + A_y^2} = 44.72 \text{ N} \quad \angle 153.4^\circ$$

$$B = 30 \text{ N} \quad \angle 90^\circ$$

b)

$$\sum F_x = 0$$

$$A_x - R_B \sin 30^\circ + 40 = 0$$

$$\sum F_y = 0$$

$$A_y + R_B \cos 30^\circ - 50 = 0$$

$$\sum M_B = 0$$

$$-0.1 \times A_y + 0.16 \times 50 - 0.1 \times 40 = 0$$

$$\therefore A_y = 20 \text{ N}$$

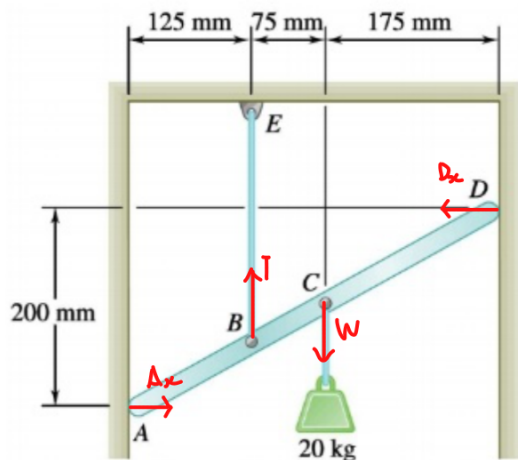
$$R_B = 34.6 \text{ N}$$

$$\therefore A = \sqrt{A_x^2 + A_y^2} = 30.2 \text{ N} \quad \angle 138.6^\circ$$

$$A_x = -22.68 \text{ N}$$

$$B = 34.6 \quad \angle 120^\circ$$

Problem 4.36: A light bar AD is suspended from a cable BE and supports a 20-kg block at C . The ends A and D of the bar are in contact with frictionless vertical walls. Determine the tension in cable BE and the reactions at A and D .



$$\sum F_x = 0$$

2020/08/29 이종민

$$A_x - D_x = 0$$

$$A_x = D_x$$

$$\sum F_y = 0$$

$$T - W = 0$$

$$T = W = 200 \times 9.81 = 1962 \text{ N}$$

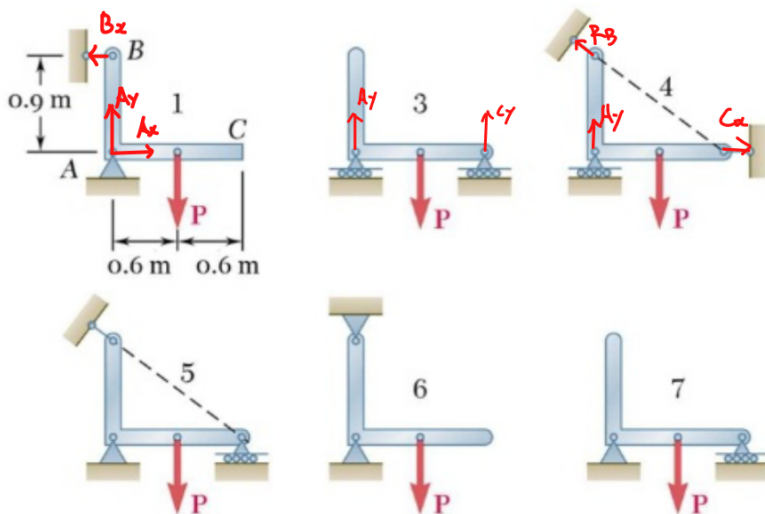
$$\sum M_A = 0$$

$$0.125 T - 0.2 W + 0.2 D_x = 0$$

$$\therefore D_x = 735.75 \text{ N}$$

$$A_x = 735.75 \text{ N}$$

Problem 4.60: The bracket ABC can be supported in the eight different ways shown. All connections consist of smooth pins, rollers, or short links. For each case, compute the reactions, assuming that the magnitude of the force P is 100 N.



Case 1.

$$\sum F_x = 0$$

$$-B_x + A_x = 0$$

$$A_x = B_x$$

$$\sum F_y = 0$$

$$A_y - P = 0$$

$$A_y = P = 100 \text{ N}$$

$$\sum M_A = 0$$

$$0.9 \times B_x - 0.6 \times P = 0$$

$$B_x = 66.6 \text{ N}$$

$$\therefore A_x = 66.6 \text{ N}$$

$$\therefore A = \sqrt{A_x^2 + A_y^2} = 120 \text{ N} \angle 56.3^\circ$$

$$B = B_x = 66.6 \text{ N} \angle 180^\circ$$

Case 3.

$$\sum F_y = 0$$

$$A_y - P + C_y = 0$$

$$A_y + C_y = P$$

$$\sum M_A = 0$$

$$-0.6 \times P + 1.2 \times C_y = 0$$

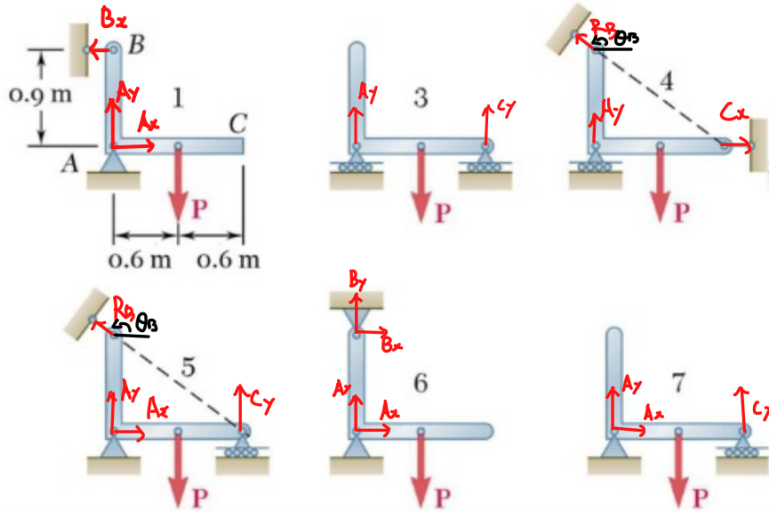
$$\therefore C_y = 50 \text{ N}$$

$$A_y = 50 \text{ N}$$

$$A = A_y = 50 \text{ N} \angle 90^\circ$$

$$C = C_y = 50 \text{ N} \angle 90^\circ$$

Problem 4.60: The bracket ABC can be supported in the eight different ways shown. All connections consist of smooth pins, rollers, or short links. For each case, compute the reactions, assuming that the magnitude of the force **P** is 100 N.



Case 4.

2020/08/04 이준원

$$\theta_B = \tan^{-1}(0.9 \div 1.2) + \frac{\pi}{2} = 126.86^\circ$$

$$\sum F_x = 0$$

$$R_B \cos \theta_B + C_x = 0$$

$$\sum F_y = R_B \sin \theta_B + A_y - P = 0$$

$$\sum M_A = -0.9 \times R_B \cos \theta_B - 0.6P = 0$$

$$B = R_B = 111.1 \text{ N} \angle 126.86^\circ$$

$$A = A_y = 11.08 \text{ N} \angle 90^\circ$$

$$C = C_x = 66.67 \text{ N} \angle 0^\circ$$

Case 5.

$$\theta_B = \tan^{-1}(0.9 \div 1.2) + \frac{\pi}{2} = 126.86^\circ$$

$$\sum F_x = R_B \cos \theta_B + A_x = 0$$

$$\sum F_y = R_B \sin \theta_B + A_y - P + C_y = 0$$

$$\sum M_A = -0.9 \times R_B \cos \theta_B - 0.6 \times P + 1.2 \times C_y = 0$$

$$\text{이때 } B \text{ 이 지름이 가해지지 않으므로 } R_B = 0$$

$$\therefore A_x = 0$$

$$C = C_y = 50 \text{ N} \angle 90^\circ$$

$$A = A_y = 50 \text{ N} \angle 90^\circ$$

Case 6.

$$\sum F_x = A_x + B_x = 0$$

$$\sum F_y = A_y + B_y - P = 0$$

$$\sum M_A = -0.9B_x - 0.6 \times P = 0$$

$$\therefore B_x = 66.66 \text{ N}$$

$$A_x = -66.66 \text{ N}$$

↓ unknowns

3 eq.)

$$A_y + B_y = 100 \leftarrow$$

Case 7.

$$\sum F_x = A_x = 0$$

$$\sum F_y = A_y - P + C_y = 0$$

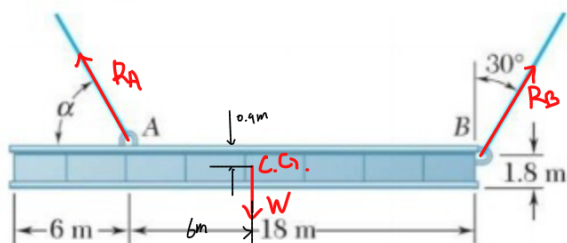
$$\sum M_A = -0.6 \times P + 1.2 \times C_y = 0$$

$$\therefore A_x = 0$$

$$C = C_y = 50 \text{ N} \angle 90^\circ$$

$$A = A_y = 50 \text{ N} \angle 90^\circ$$

Problem 4.64: A uniform plate girder weighing 12 kN is held in a horizontal position by two crane cables. Determine the angle α and the tension in each cable.



$$\sum F_x = -R_A \cos \alpha + R_B \sin 30^\circ = 0$$

$$\sum F_y = R_A \sin \alpha + R_B \cos 30^\circ - W = 0$$

$$\sum M_A = -6 \times W + 18 \times R_B \cos 30^\circ = 0$$

$$\therefore R_B = 4.618 \text{ kN}$$

$$\therefore R_A \cos \alpha = 2.309 \text{ kN}$$

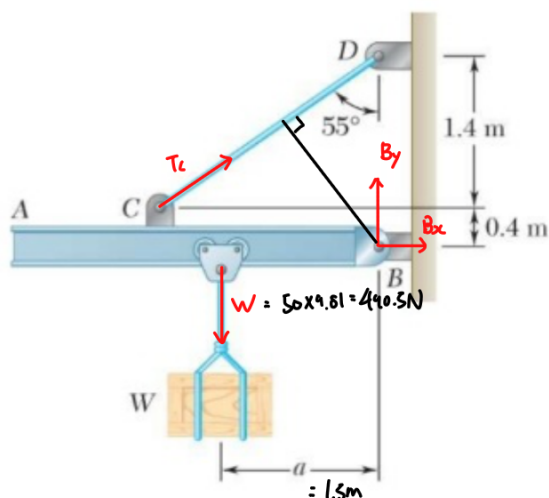
$$R_A \sin \alpha = 8.000 \text{ kN}$$

$$R_A = \sqrt{(R_A \cos \alpha)^2 + (R_A \sin \alpha)^2} = 8.327$$

$$\alpha = 73.9^\circ$$

2020/08/09 이준현

Problem 4.69: A 50-kg crate is attached to the trolley-beam system shown. Knowing that $a = 1.5$ m, determine (a) the tension in cable CD, (b) the reaction at B.



$$\sum F_x = T_C \cos 55^\circ + B_x = 0$$

$$\sum F_y = T_C \sin 55^\circ - W + B_y = 0$$

$$\sum M_B = -1.5 \sin 55^\circ \times T_C + 1.5W = 0$$

$$\therefore T_C = 498.99 \text{ N}$$

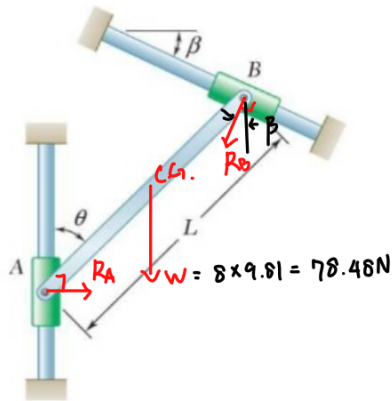
$$\therefore B_y = 204.29 \text{ N}$$

$$B_x = -406.74$$

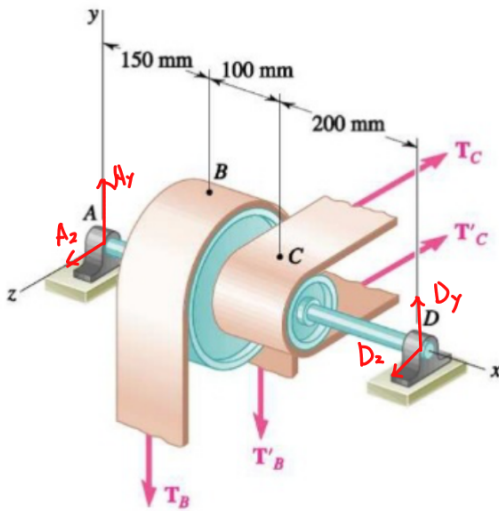
$$T_C = 498.99 \text{ N } \angle 35^\circ$$

$$B = \sqrt{B_x^2 + B_y^2} = 456.9 \text{ N } \angle 153.4^\circ$$

Problem 4.85: An 8-kg slender rod of length L is attached to collars that can slide freely along the guides shown. Knowing that the rod is in equilibrium and that $\beta = 30^\circ$, determine (a) the angle θ that the rod forms with the vertical, (b) the reactions at A and B.



Problem 4.91: Two transmission belts pass over a double-sheaved pulley that is attached to an axle supported by bearings at A and D. The radius of the inner sheave is 125 mm and the radius of the outer sheave is 250 mm. Knowing that when the system is at rest, the tension is 90 N in both portions of belt B and 150 N in both portions of belt C, determine the reactions at A and D. Assume that the bearing at D does not exert any axial thrust.



2020/08/09 03:21

$$\sum F_x = A_x - R_B \sin \beta = 0$$

$$\sum F_y = -W - R_B \cos \beta = 0$$

$$\sum M_B = L \cos \theta \times R_A + \frac{L \sin \theta}{2} \times W = 0$$

$$\therefore R_B = 90.62 \text{ N} \quad \angle 60^\circ$$

$$R_A = 45.31 \text{ N} \quad \angle 180^\circ$$

$$L R_A + \frac{L}{2} W \tan \theta = 0$$

$$\tan \theta = -\frac{2 R_A}{W}$$

$$\theta = \tan^{-1} \left(-\frac{2 R_A}{W} \right)$$

$$a) = 41.1^\circ$$

$$\sum F_x = 0$$

$$\sum F_y = A_y - 2T_B + D_y = 0$$

$$\sum F_z = A_z - 2T_C + D_z = 0$$

$$\sum M_{A,x} = 0$$

$$\sum M_{A,y} = 0.25 \times 2T_C - 0.45 D_z = 0$$

$$\sum M_{A,z} = -0.15 \times 2T_B + 0.45 D_y = 0$$

$$\therefore D_z = 166.6 \text{ N}$$

$$D_y = 60 \text{ N}$$

$$A_z = 133.4 \text{ N}$$

$$A_y = 20 \text{ N}$$

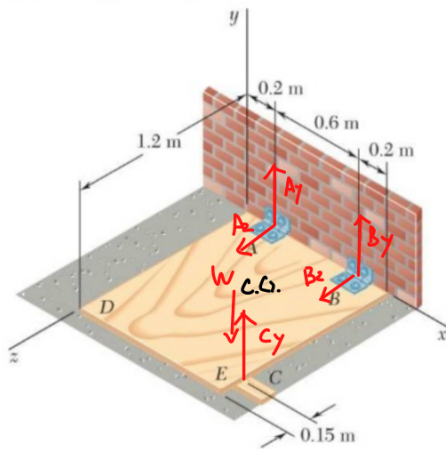
$$T_B = 90 \text{ N}$$

$$T_C = 150 \text{ N}$$

$$\text{Ans. } \underline{A} = (20 \text{ N}) \hat{j} + (133.4 \text{ N}) \hat{k}$$

$$\underline{D} = (60 \text{ N}) \hat{j} + (166.6 \text{ N}) \hat{k}$$

Problem 4.99: An opening in a floor is covered by a 1×1.2 -m sheet of plywood of mass 18 kg . The sheet is hinged at A and B and is maintained in a position slightly above the floor by a small block C . Determine the vertical component of the reaction (a) at A , (b) at B , (c) at C .



$$W = 18 \times 9.81 = 176.58 \text{ N}$$

2020/08/09 이종원

$$\sum F_y = A_y + B_y - W + C_y = 0$$

$$\sum F_z = A_z + B_z = 0$$

$$\underline{r}_{WA} = (0.3, 0, 0.6)$$

$$\underline{W} = (0, -176.58, 0)$$

$$\underline{r}_{BA} = (0.6, 0, 0)$$

$$\underline{F}_B = (0, B_y, B_z)$$

$$\underline{r}_{CA} = (0.8, 0, 1.05)$$

$$\underline{F}_C = (0, C_y, 0)$$

$$\sum M_A = \underline{r}_{WA} \times \underline{W} + \underline{r}_{BA} \times \underline{F}_B + \underline{r}_{CA} \times \underline{F}_C = 0$$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0.3 & 0 & 0.6 \\ 0 & -176.58 & 0 \end{vmatrix} + \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0.6 & 0 & 0 \\ 0 & B_y & B_z \end{vmatrix} + \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0.8 & 0 & 1.05 \\ 0 & C_y & 0 \end{vmatrix} = 0$$

$$\therefore (105.948)\hat{i} + (-52.974)\hat{k} + (0.6B_z)\hat{j} + (0.6B_y)\hat{k} + (-1.05C_y)\hat{i} + (0.8C_y)\hat{k} = 0$$

정리하면

$$\therefore C_y = 100.9 \text{ N}$$

$$B_z = 0$$

$$B_y = -46.2 \text{ N}$$

$$\therefore A_y = 121.9 \text{ N}$$

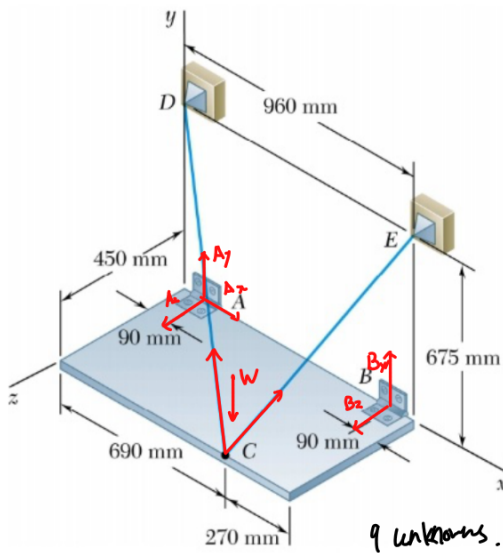
$$A_z = 0$$

$$a) \underline{A} = (121.9 \text{ N})\hat{j}$$

$$b) \underline{B} = (-46.2 \text{ N})\hat{j}$$

$$c) \underline{C} = (100.9 \text{ N})\hat{j}$$

Problem 4.117: A 100-kg uniform rectangular plate is supported in the position shown by hinges A and B and by cable DCE that passes over a frictionless hook at C . Assuming that the tension is the same in both parts of the cable, determine (a) the tension in the cable, (b) the reactions at A and B . Assume that the hinge at B does not exert any axial thrust.



9 unknowns.

$$\vec{CD} = (-0.69, 0.675, -0.45) \text{ m}$$

$$|\vec{CD}| = \sqrt{(-0.69)^2 + (0.675)^2 + (-0.45)^2} = 1.065 \text{ m}$$

$$\vec{T}_{CD} = T \frac{\vec{CD}}{|\vec{CD}|} = T(-0.6478, 0.6338, -0.4225) \text{ N}$$

$$\vec{CE} = (0.27, 0.675, -0.45) \text{ m}$$

$$|\vec{CE}| = \sqrt{(0.27)^2 + (0.675)^2 + (-0.45)^2} = 0.855 \text{ m}$$

$$\vec{T}_{CE} = T \frac{\vec{CE}}{|\vec{CE}|} = T(0.3157, 0.7894, -0.5263)$$

$$\vec{r}_{CA} = (0.6, 0, 0.45) \text{ m}$$

$$\vec{r}_{BA} = (0.78, 0, 0) \text{ m}$$

$$\vec{r}_W = (0.39, 0, 0.225) \text{ m}$$

$$\vec{W} = (0, -981, 0) \text{ N}$$

$$\sum F_x = A_x - 0.6478T + 0.3157T = 0$$

$$\sum F_y = A_y + B_y - W + 0.6338T + 0.7894T = 0$$

$$\sum F_z = A_z + B_z - 0.4225T - 0.5263T = 0$$

$$\sum M_A = \vec{r}_{CA} \times \vec{T}_{CD} + \vec{r}_{BA} \times \vec{T}_{CE} + \vec{r}_{BA} \times (B_y \hat{j} + B_z \hat{k}) + \vec{r}_W \times \vec{W} = 0$$

$$T \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0.6 & 0 & 0.45 \\ -0.6478 & 0.6338 & -0.4225 \end{vmatrix} + T \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0.6 & 0 & 0.45 \\ 0.3157 & 0.7894 & -0.5263 \end{vmatrix} + \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & B_y & B_z \end{vmatrix} + \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0.39 & 0 & 0.225 \\ 0 & -981 & 0 \end{vmatrix} = 0$$

$$\hat{i}: (-0.2852T - 0.3552T + 220.7) = 0$$

$$\text{a) } \therefore T = 345 \text{ N}$$

$$\hat{j}: -(0.0380T - 0.4578T + 0.7894T) = 0$$

$$\therefore B_z = 185 \text{ N}$$

$$\hat{k}: (0.3802T + 0.4136T + 0.78B_y - 382.5) = 0$$

$$\therefore B_y = 112 \text{ N}$$

$$\therefore A_x = 114 \text{ N}$$

$$A_y = 377 \text{ N}$$

$$A_z = 142 \text{ N}$$

$$\text{b) } \therefore \vec{A} = (114 \text{ N})\hat{i} + (377 \text{ N})\hat{j} + (142 \text{ N})\hat{k}$$

$$\vec{B} = (112 \text{ N})\hat{j} + (185 \text{ N})\hat{k}$$